Shengxu Mu

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EDUCATION

- **Tsinghua University** Beijing, China
  B.Eng. in Computer Science & Technology *08/2024 – Present*
  B.Econ. in Economics & Finance (Dual Degree)
  Department of Computer Science and Technology
  GPA: **3.8/4.0**

Core Courses: Software Engineering (A+) / Digital Logic Design (A+) / Deep Learning and Financial Data Analysis (A+) / Programing and Training (A) / Econometrics (A) / Applications of AI in Quantitative Finance (A-)

RESEARCH INTERESTS

My research interests center on **human-centered artificial intelligence**, with a focus on how intelligent agents understand human intentions and adapt to evolving task contexts. Building on my research in human–AI collaboration, I am extending my work toward **agent memory**, **context management**, and **long-horizon interactions**. By integrating human-centered perspectives with algorithmic and systems approaches, I aim to develop reliable, efficient, and adaptive agents that support sustained collaboration between humans and AI. I also explore **machine learning for quantitative finance**, with a focus on equity return prediction and financial market simulation.

PUBLICATIONS

Cache Engineering for Agents

Tianyu Fu, Ruoxi Wang*, Zihan Min*, Zhongqi Ren, Zhihao Li, Tengxuan Liu, Shengxu Mu, Hanling Zhang, Yu Wang*
arXiv preprint, 2026. [arxiv] [code] [website]

MANUSCRIPTS UNDER REVIEW

ACM CHI 2027 *Two manuscripts under review*

- Negotiable AI participation in human–AI collaborative writing.
- Task representations and sensemaking in AI-assisted news writing.

Research topics are summarized to preserve anonymity during peer review.

RESEARCH EXPERIENCE

- **Tsinghua University, Pervasive Human Computer Interaction Lab** Beijing, China
  PI Lab *10/2025 – Present*
  Advisor: Prof. Chun Yu
  - **Human–AI collaboration and co-creation**, exploring how interactive AI systems can support human agency, shared understanding, and adaptive collaboration in complex creative tasks.
  - Investigated **negotiable boundaries of AI participation in human–AI co-writing**, studying how humans and AI dynamically coordinate roles, permissions, and control. Designed and evaluated interaction mechanisms that make these boundaries explicit and adjustable.
  - Explored **shared task representations for AI-assisted news writing**, investigating how externalizing task goals, information, and writing processes can support shared understanding and effective human–AI coordination.

- **Tsinghua University, Nanoscale Integrated Circuits and System Lab, Energy Efficient Computing Group** Beijing, China
  NICS-EFC Lab *07/2026 – Present*
  Advisor: Prof. Yu Wang
  - **Efficient agent systems focusing on new paradigms for context management**.
  - Explore *Cache Engineering for Agents*, contributing to its algorithm design. This new direction enables agents to manage their own KV caches and iteratively improve cache policies, with the accompanying *AKV* serving system supporting efficient reuse of retained states for long-horizon execution.

QUANTITATIVE FINANCE PROJECTS

- **Residual StockMixer for Equity Return Ranking**
  - Developed a **residual factor-token extension to StockMixer** for cross-sectional equity return prediction, learning latent cross-stock dependencies to refine return rankings.
  - Achieved a test RankIC of **0.0237**, compared with **0.0149** for the StockMixer baseline, and evaluated robustness across data configurations and random seeds.
- **Multi-Asset Limit Order Book Generation**
  - Extended **diffusion-based market simulation** to multiple assets, investigating parameter sharing, cross-asset graph coupling, and no-arbitrage guidance.
  - Identified the benefits of shared representations and limitations of explicit asset coupling. Improved **autoregressive rollout stability**, reducing spot–perpetual basis Wasserstein distance by approximately **46–90**× through decode-time interventions.
- **Fundamental Quality Factor and Backtesting**
  - Constructed an **A-share financial quality factor** integrating profitability, operating cash flow, and leverage, with announcement-date alignment and cross-sectional normalization.
  - Evaluated factor effectiveness through **IC/RankIC analysis and decile portfolios**, and implemented a long-only backtesting framework incorporating transaction costs and price-limit constraints.

LEADERSHIP & ACTIVITIES

- **Tsinghua University Student Union** 09/2024 – 08/2026
  Deputy Director, Arts Department
  - Organized and coordinated campus-wide cultural events and performances, managing cross-team collaboration, event planning, and on-site execution.
- **Student Union, Department of Computer Science & Technology** 03/2026 – Present
  Deputy Director, Arts and Publicity Department
  - Coordinated departmental cultural activities and publicity initiatives, supporting student engagement, communications, and collaboration across teams.
- **Hong Kong–Singapore Overseas Field Study Delegation** 07/2026
  Leader, Department of Computer Science & Technology
  - Led an overseas field study on AI applications, evaluation, and governance, coordinating institutional outreach, academic and industry visits, and research planning.

AWARDS

- 2025: **Student Leadership and Service Excellence Scholarship**, Department of Computer Science & Technology, Tsinghua University

- 2025: **Volunteer and Community Service Excellence Scholarship**, Department of Computer Science & Technology, Tsinghua University
- 2024: **Friends of Tsinghua – Cyrus Tang Moral Education Scholarship**, Tsinghua University

SKILLS

- **Programming & tools:** Python, C++, C, Java, Bash, Git, LaTeX, SystemVerilog
- **Deep Learning Package:** PyTorch, Transformers